

Falcon 2.0T low system voltage from failed 180A alternator

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Area: bulletins

1 Condition

Some Falcon 2.0T vehicles may display a battery or charging warning lamp accompanied by dimming lights and reduced electrical performance. When this occurs, DTC P0562 (System Voltage Low) is stored in the powertrain control module. The condition is often more pronounced under high electrical load such as headlamps and blower operating together. System behavior is described in the Falcon Charging Service Manual :: Charging System Overview .

2 Cause

This condition is caused by degraded output from the 180A alternator, part number ALT-8810, which fails to maintain regulated system voltage under load. As measured charging voltage falls below the calibrated threshold, the PCM sets DTC P0562. Worn brushes or a failed internal rectifier within ALT-8810 are the typical failure modes. Component details are provided in the Falcon Charging Service Manual :: Alternator , and the code is documented in the Falcon Charging Service Manual :: System Voltage Codes .

3 Diagnostic Procedure

Confirm DTC P0562 is stored, then perform a charging system load test following PROC-CHG-DIAG. Measure alternator output at idle and at elevated RPM under load, comparing against the specification; output that falls short with good battery and cable condition confirms a failed alternator. Inspect the generator control circuit before condemning the unit, as covered in the Falcon Charging Service Manual :: Generator Control Circuit Code . The complete test sequence is given in the Falcon Charging Service Manual :: Charging System Diagnosis Procedure .

4 Repair Procedure

Replace the alternator ALT-8810 following PROC-ALT, ensuring the drive belt and tensioner are inspected during reinstallation. After installation, recharge or verify the battery state of charge, clear DTC P0562, and confirm regulated system voltage under load. Removal and torque specifications are detailed in the Falcon Charging Service Manual :: Alternator Replacement Procedure , within the Falcon Charging Service Manual :: Alternator Sub-System Service .